

Network level shifts in E/I balance during chronic epilepsy revealed by aperiodic EEG dynamics

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Project page



Poster made with my Claude skill

Abstract

Chronic seizures progressively reshape brain-wide connectivity and excitation–inhibition (E/I) balance, but medication effects make these dynamics difficult to study in patients. Using a chronic epileptic Lafora disease (Epm2a^{−/−}) mouse model, we tracked how network-level E/I balance evolves across epileptogenesis and ictogenesis using the aperiodic exponent, a noninvasive EEG proxy for E/I ratio and beta-band oscillatory metrics. Laforin deficient (Epm2a^{−/−}) Lafora disease mice exhibited steeper baseline aperiodic slopes and elevated beta activity, indicating increased tonic inhibition. Isoflurane, a GABA-A receptor agonist, produced stronger spectral slope steepening in wild-type animals than in knockouts, reflecting altered GABAergic tone in Epm2a^{−/−} mice. During chemically induced seizures, aperiodic measures showed systematic variation across pre-ictal, ictal, and post-ictal periods, revealing phase-specific shifts between inhibitory and excitatory dominance. These results suggest that cortical E/I balance varies dynamically across seizure phases and severity. By resolving these transitions in real time, aperiodic EEG features offer a promising biomarker for seizure staging and vulnerability assessment. This work provides mechanistic insight into network-level adaptations in chronic epilepsy while avoiding pharmacological confounds.

Method

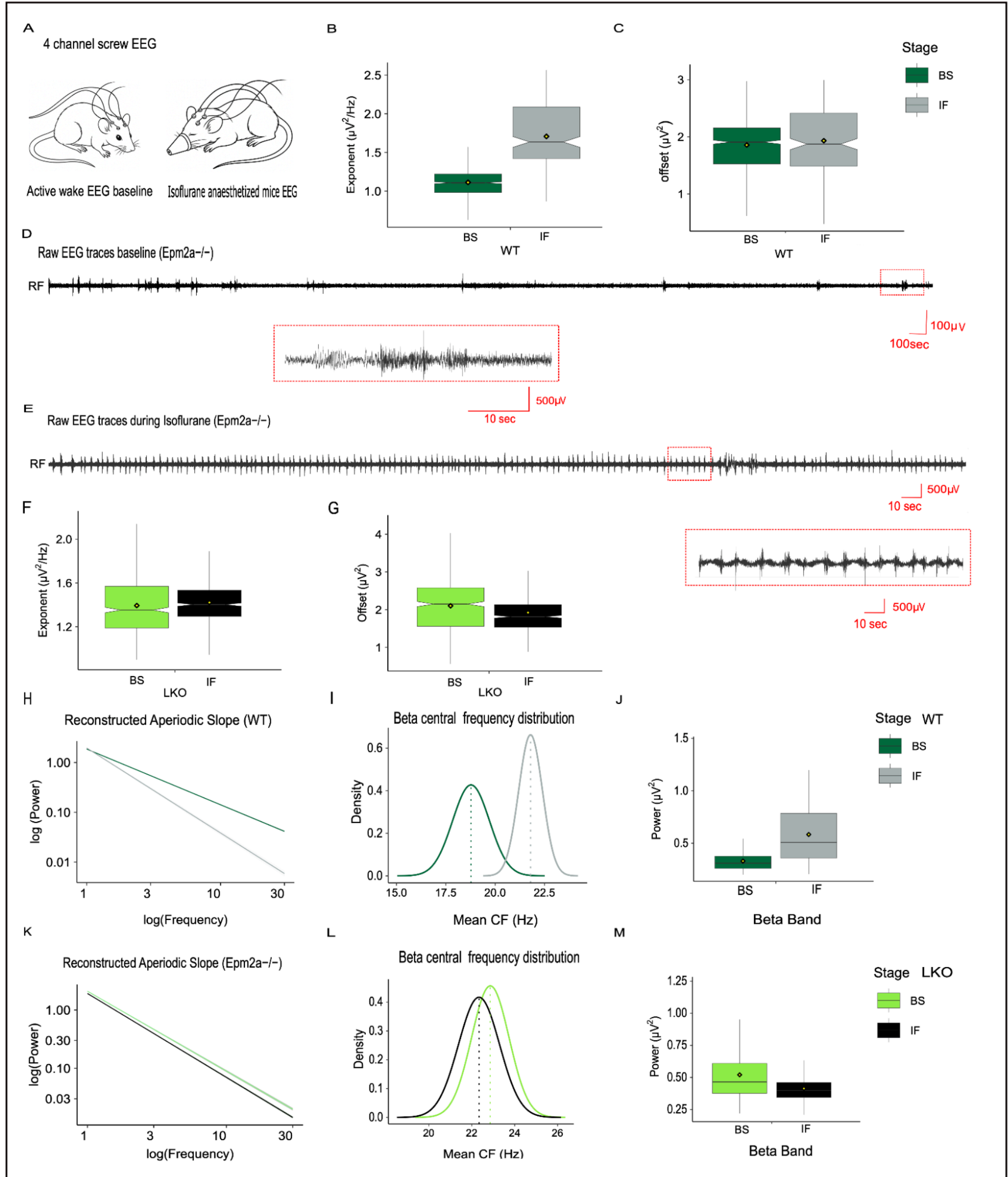


Figure 1: E/I balance and aperiodic exponents. Tracking network-level E/I balance across epileptogenesis and ictogenesis using the aperiodic exponent. Using a chronic epileptic Lafora disease (Epm2a^{−/−}) mouse model, we tracked how network-level E/I balance evolves across epileptogenesis and ictogenesis using the aperiodic exponent, a noninvasive EEG proxy for E/I ratio and beta-band oscillatory metrics.

Results

Laforin deficient Lafora disease mice exhibited steeper baseline aperiodic slopes and elevated beta activity, indicating increased tonic inhibition. Isoflurane produced stronger spectral slope steepening in wild-type animals. Aperiodic measures showed systematic variation across pre-ictal, ictal, and post-ictal periods.

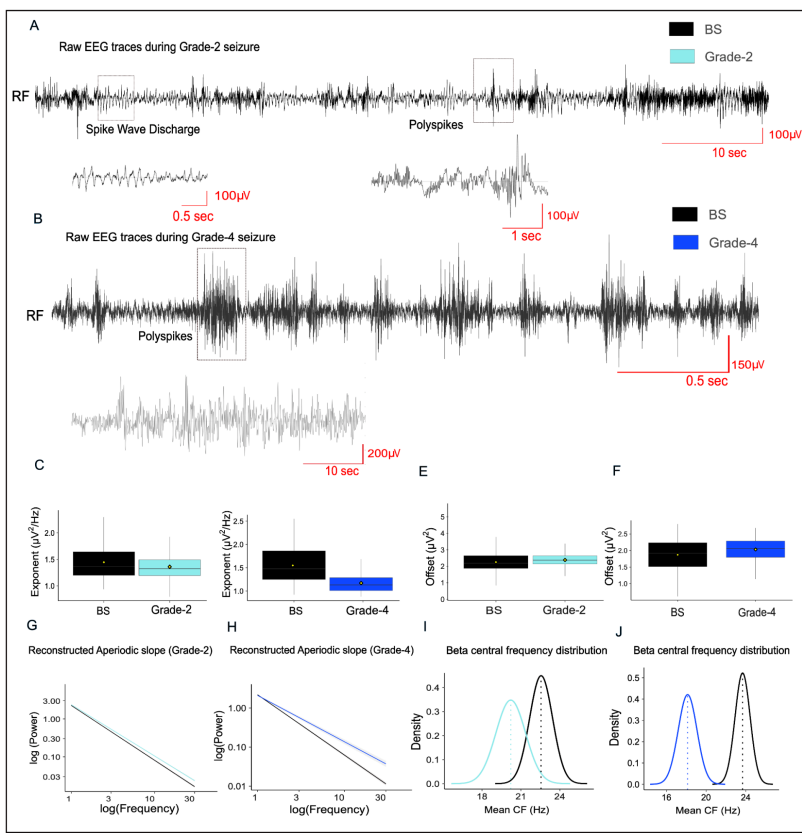


Figure 2: Aperiodic slopes and beta activity. Baseline characteristics in Laforin deficient mice compared to wild-type.

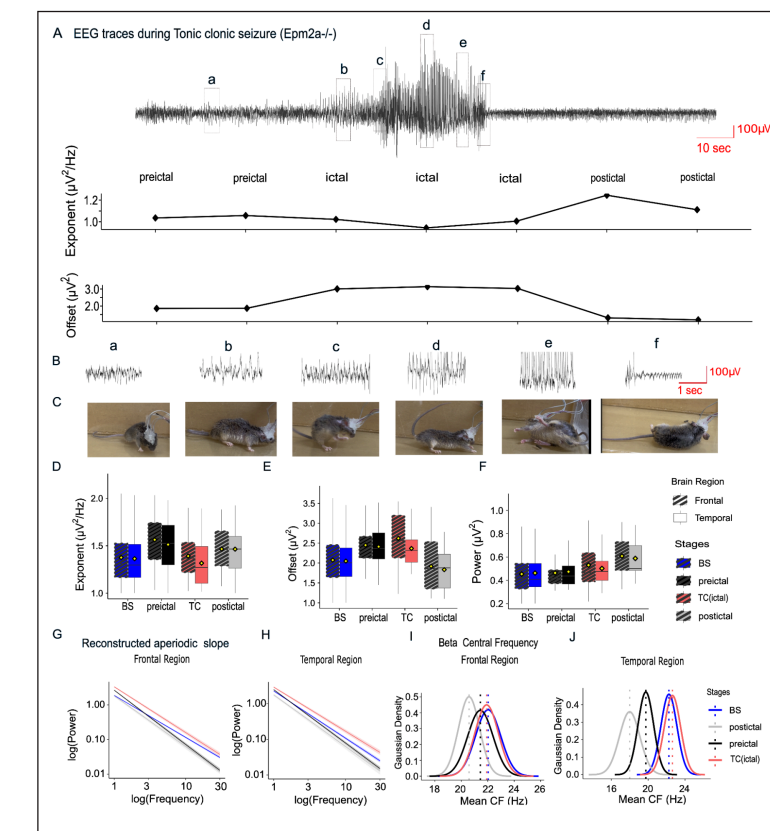


Figure 3: Pharmacological effects. Effects of Isoflurane on spectral slope steepening in wild-type vs knockouts.

Introduction

Epilepsy is a dynamic network disorder in which recurrent seizures progressively reshape brain-wide structure, connectivity, and the balance between excitation and inhibition (E/I). Repeated seizures trigger long-lasting neuroplastic changes (Wang et al., 2025), destabilize local and global functional organization (Barrios-Martinez et al., 2025), and increase vulnerability to subsequent events (Liang et al., 2020). Disruptions in E/I balance are considered a central hallmark of epileptogenesis (Liu et al., 2023), reflected in alterations in the Hurst exponent in chronic temporal lobe epilepsy (Xie et al., 2024), dynamic E/I fluctuations across seizure phases (Žiburkus et al., 2013), and seizure-induced strengthening of excitatory synapses via BDNF signalling (Nasrallah et al., 2022). However, how E/I balance evolves across seizure stages and grades remains poorly understood, particularly in humans, where ethical constraints and anti-epileptic drugs (AEDs) obscure intrinsic circuit dy...

Conclusion

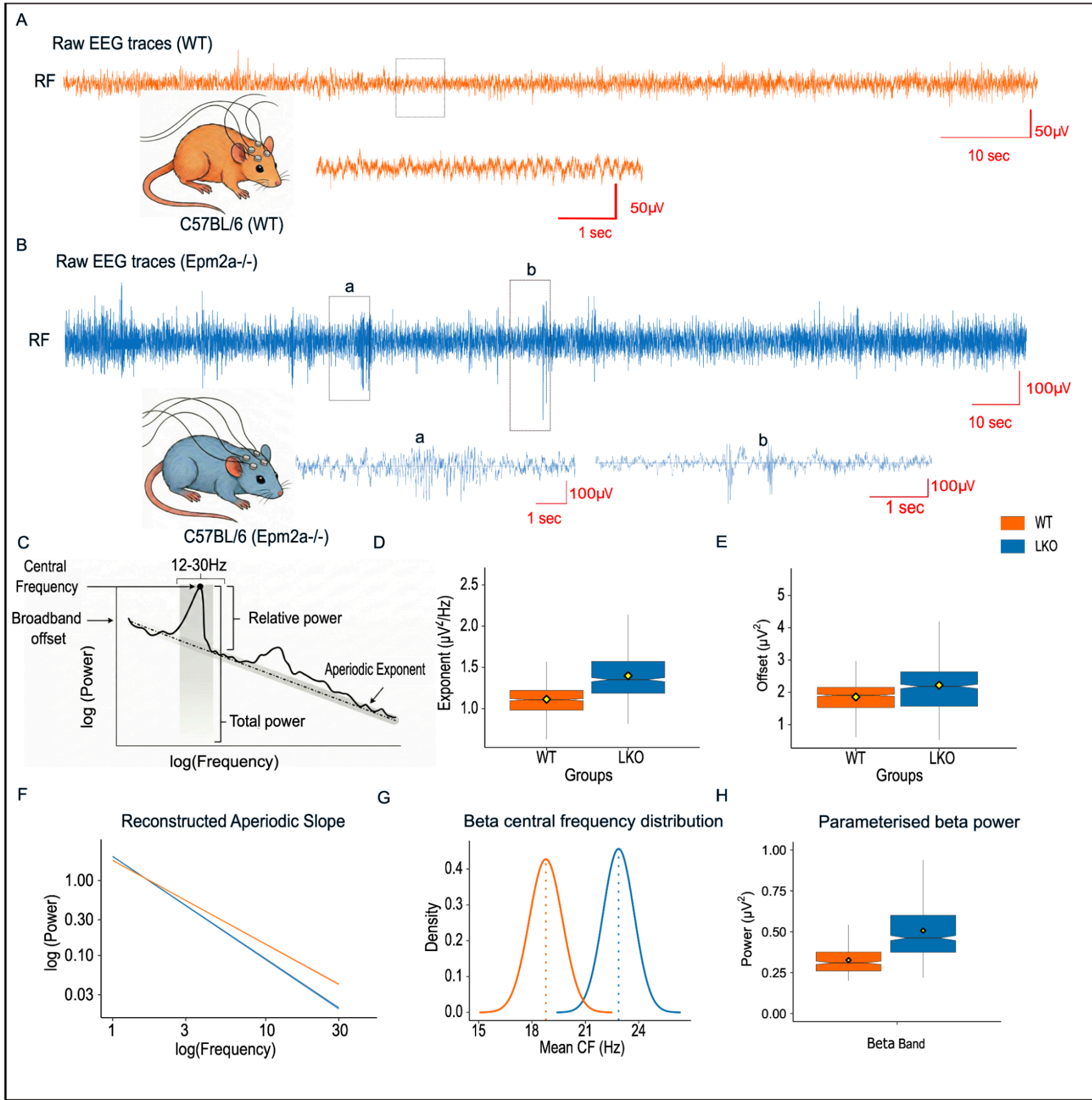


Figure 4: Seizure phases. Systematic variation of aperiodic measures across pre-ictal, ictal, and post-ictal periods. These results suggest that cortical E/I balance varies dynamically across seizure phases and severity. By resolving these transitions in real time, aperiodic EEG features offer a promising biomarker for seizure staging and vulnerability assessment.